

GABRIEL MILANEZ CAVALHERI

Mogi Guaçu, SP, Brazil | (19) 98222-1223 | gmcavalheri@gmail.com
linkedin.com/in/gabriel-milanez-cavalheri | github.com/GMCavalheri

SUMMARY

Physicist (UFSCar) pursuing an MBA in Artificial Intelligence and Big Data (USP), with hands-on training in data science and data engineering. Experience building self-directed projects covering ETL pipelines, Data Lake and Data Warehouse integration, Machine Learning modeling, and Docker-based automation. Seeking a position as a Data Engineer, Data Scientist, or Data Analyst to apply technical skills to data-driven solutions.

EDUCATION

- **MBA in Artificial Intelligence and Big Data** — Universidade de São Paulo (USP) | 07/2025 – 10/2026 (in progress)
- **Bachelor's Degree in Physics** — Universidade Federal de São Carlos (UFSCar) | Completed 03/2025
- **Undergraduate Research in Applied Acoustics** — UFSCar | 2023 – 2025

DATA PROJECTS

Data Warehouse and Data Lake Integration with ETL Pipeline

<https://github.com/GMCavalheri/Complete-BI-and-ETL-with-Data-Warehouse-and-Datalake>

- Stack: Python, PostgreSQL, PyArrow, DuckDB, Polars, Docker, SQL
- Built an ETL pipeline integrating a Data Lake with a relational Data Warehouse, supporting SQL queries from basic to advanced; entire environment containerized with Docker for reproducibility by other users.

Currency Exchange ETL Pipeline Orchestrated with Airflow

<https://github.com/GMCavalheri/Complete-ETL-Pipeline-with-Airflow>

- Stack: Python, PostgreSQL, Apache Airflow, Docker
- ETL pipeline that consumes an API of world currency exchange rates against the US dollar, with automatic daily updates at the same time, orchestrated with Airflow and served via a dashboard; containerized with Docker.

Pokédex Inteligente — Web Application with Django and LLM

- Stack: Python, Django, PokéAPI, LLM integration (OpenRouter)
- Django web application that consumes data from the PokéAPI and enables natural-language queries over the data through integration with a language model (LLM).

Other Applied Projects

- **Diabetes prediction** on a patient dataset using Random Forest.
- **Mechanical gear fault diagnosis** from vibration signals, using Deep Neural Networks.
- **Brain tumor detection** in MRI scans using Convolutional Neural Networks.
- **Customer sentiment analysis** on product reviews using Natural Language Processing techniques.
- **Movie recommendation system** using the APRIORI algorithm on a large dataset.
- **Legal entity recognition** in legal documents using a Natural Language Processing model.
- **Geospatial analysis** of software developer distribution.

TECHNICAL SKILLS

- **Data & Engineering:** SQL (PostgreSQL, MySQL, Spark SQL), NoSQL (MongoDB), Apache Spark, Apache Airflow, ETL, PyArrow, DuckDB, Polars
- **Languages:** Python (Pandas, NumPy, SciPy, Scikit-Learn, TensorFlow, Keras, Streamlit), R (tidyverse, dplyr, ggplot2)
- **Machine Learning & AI:** Supervised, Unsupervised, and Reinforcement Learning; Neural Networks (MLP, CNN, RNN, Transformers, Generative); NLP (NLTK, spaCy)
- **Cloud & DevOps:** AWS, Docker, Kubernetes (basic), Git/GitHub
- **Analytics & Visualization:** Microsoft Excel, Power BI, Seaborn, applied statistics (inference, hypothesis testing, distributions)
- **Generative AI & Automation:** Claude, ChatGPT, Gemini, Ollama and local open-source models; n8n, Make
- **Methodology:** Agile, Kanban, Clean Code, Data Storytelling

LANGUAGES

- Portuguese — Native
- English — Advanced (C1 Advanced)
- Spanish — Basic